# Purdue Electric Racing Suspension Test Rig

PURDUE UNIVERSITY
Mechanical Engineering

**Student Names: Trevor Kates, Hayden Hohman, Arthur Donohue, Tanner Erny, Jacob Lecure, Stephen Sinning**
**Faculty Advisor: Professor Jingang Li**

**Abstract:**
Purdue Electric Racing (PER) currently lacks a validation apparatus for testing suspension performance prior to on-track use. As a result, the team is forced to design the suspension conservatively, reducing its overall efficiency. This project seeks to develop a static quarter-car suspension testing rig that simulates cornering forces to evaluate suspension member material behavior and support stronger data-driven decisions. This rig will provide the team with insight into suspension member deflection, member loading, and wheel dynamics under intense loading.

## Introduction

- The PER team utilizes a suspension system with five primary load-carrying members per wheel
- During cornering, the outside wheels experience large normal and lateral loads
- These loads cause suspension members to undergo significant compression and bending
- Material selection and design decisions must be made prior to assembly without full validation data
- This project develops a method to test suspension materials and configurations under realistic loading conditions
- **Novelty:** Rig simultaneously loads wheel hub normally and laterally with steering adjustability

## Key Requirements

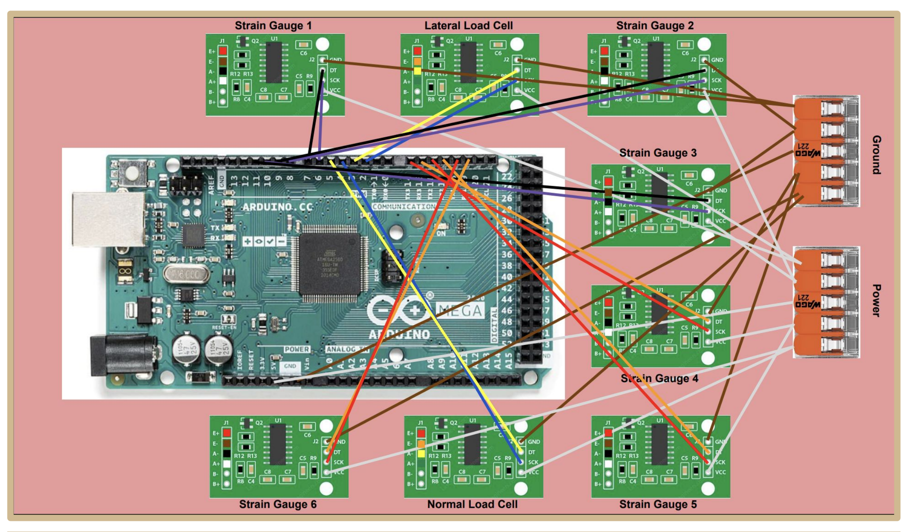
- Apply up to **4000 newtons** of force laterally and **1000 newtons** of force normally at the same time
- Allow for steering adjustability of **+/- 15 degrees**
- Ensure accurate and calibrated sensor data
- Operate without safety concerns
- Construct rig within the **$1000** budget

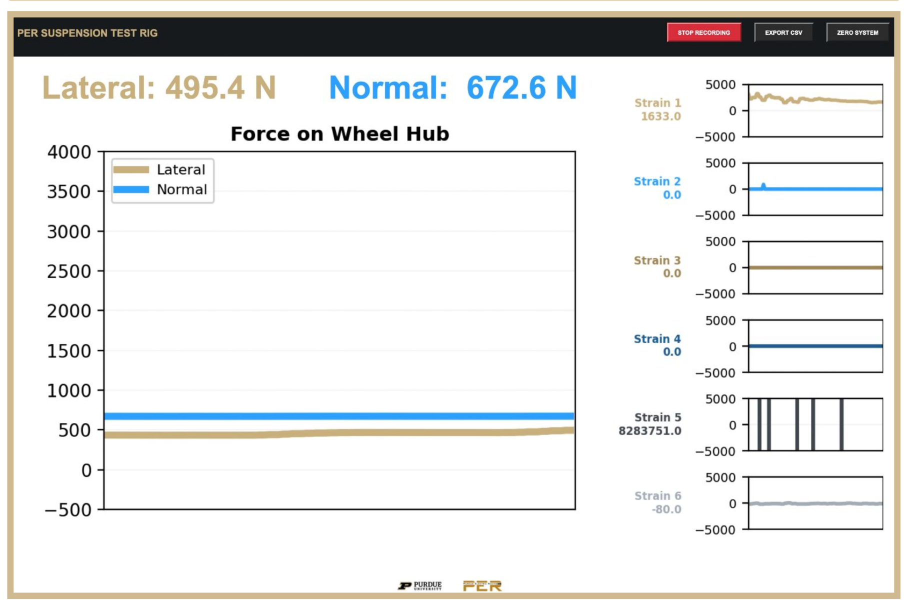
## Sensors & Data Collection

**Load Cells** - calibrated in ME32301 Lab with Universal Testing Machine

**Strain Gauges -** wheatstone bridge setup to eliminate temperature variability and bending moments

**User Interface -** sensor voltage data amplified by HX711 chips and sent from Arduino to custom live-monitor Python script UI



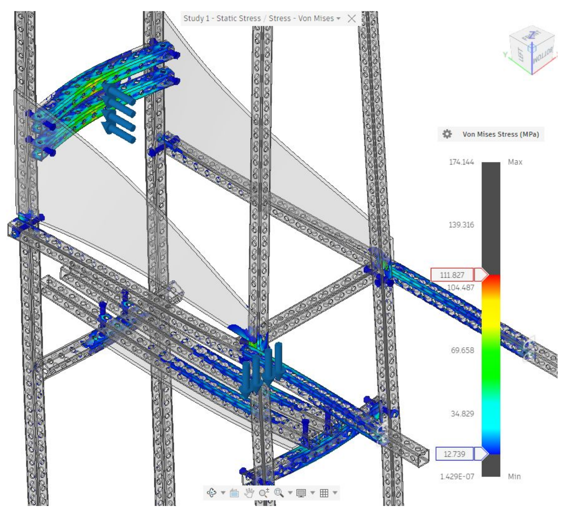

## Mechanical Design

- The frame is made from ⅛" black iron tubing
- The frame is bolted together with 5/16" bolts
- 80-20 aluminum squared the frame
- Actuators are connected to frame using clevis tabs and ball joints

- The pumps are connected via tubing to the **hydraulic actuators**
- The hydraulic pumps are manually pumped to create pressure
- The pressure pushes the hydraulic fluid through the tubing and into the actuators

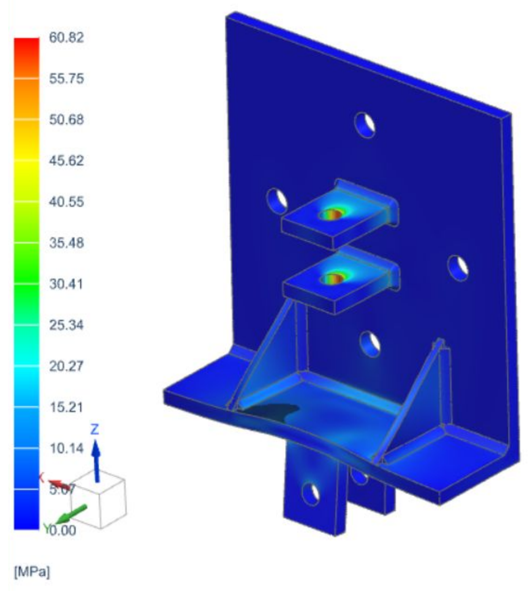
- Two total actuators - one in a normal position to wheel hub and the other is in a lateral position
- Fluid moves into the actuators and creates force on the hub
- Force is **released** by draining the fluid from the actuator back into the pump

## Design Validation



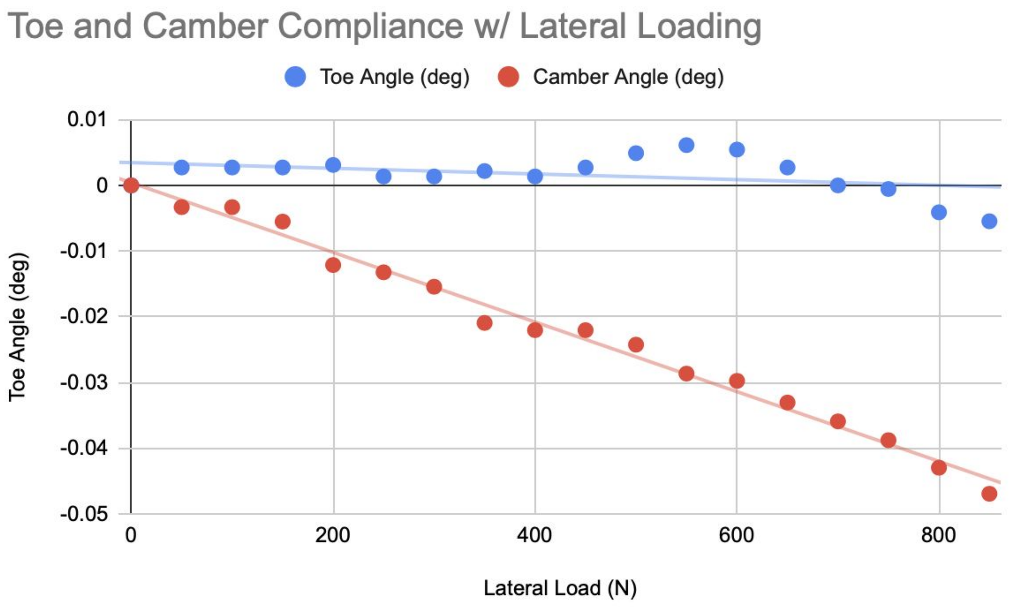
- 6 mm mesh w/ local adaptive mesh refinement
- **F.S. > 2.1**
- Max displacement < 0.23 mm
- 5x thicker tubes used in prototype to increase safety.

- 1.5 mm mesh size
- **F.S. > 3.3**
- Max displacement < 0.012 mm




## Results

- Load was applied by the **actuators** into the wheel hub in the lateral and normal directions
- **Dial indicators** were attached to a plate that allowed for the measurement of toe and camber angles
- Lateral and normal force was applied incrementally over specified load range from stakeholder
- Toe and Camber deflection values were recorded at each increment and post-processed to form a graphs showing the deflection angle against load.




## Conclusion

- Load was applied to the car through the hub in a lateral and normal direction creating **stress** and **strain** on the **suspension members**
- Dial indicators allowed for the validation of both toe and camber angles of the hub
- Data **validated** that **carbon fiber** suspension members **outperformed** their steel counterparts by wide margins on all fronts
- The data also validated the simulation models which allows for a **higher factor of safety**
- The force required for the preload stage of the suspension members was discovered
- Testing **revealed a leak** in the plate adapter on the hub that connects to the motor, which otherwise would not have been found
- Also **revealed a conduction spot** on the coating that was applied to the suspension members
- Next steps are to create a standard operating procedure that can be easily understood and followed by anyone
- Final step is to train some members of the Purdue Electric Racing team on how to use the rig and how to set it up

# *TESTING THE SUSPENSION SYSTEM ON A CAR WITHOUT DRIVING IT*